

GIA VALDEZ

MS student at DePaul University
Chicago, Illinois, United States

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EDUCATION

DePaul University, Chicago IL

M.S. in Biological Sciences

11.2025

Thesis Advisor: Dr. Margaret Bell

Thesis: "Early-Life Exposure to PFAS Affects Adolescent Neuroimmune Activity"

DePaul University Honors College

06.2023

D.M -- B.S. in Neuroscience and Psychology

Minors in Biology and Japanese Language

RESEARCH EXPERIENCE

Lab Manager

07.2025–Present

PI: Dr. Margaret Bell; Dept of Biological Sciences, DePaul University, Chicago, Illinois

Research focus: Effects of early life perinatal polychlorinated biphenyls (PCB) on neonatal and adolescent neuroimmune and neuroendocrine endpoints.

Techniques: In vivo dosing/exposure, perfusion, multi-organ tissue collection, primary cell culture, MACS cell separation, RNA isolation, cDNA synthesis, qPCR, ICC, confocal microscopy, ImageJ, statistics, R/Python/Prism data analysis.

Graduate Research Assistant

09.2024–Present

PI: Dr. Margaret Bell; Dept of Biological Sciences, DePaul University; Chicago, Illinois

MS Thesis: Effects of early-life PFOS exposure on neural and neuroimmune responses after an inflammatory challenge during adolescence.

Techniques: In vivo chemical exposures, targeted tissue collection, MACs cell separation, RNA isolation, qPCR, RNAseq data processing/interpretation, analysis and figure prep using Jupyter Notebook and Python.

Undergraduate Research Assistant

03.2022–08.2024

PI: Dr. Margaret Bell; Dept of Biological Sciences, DePaul University; Chicago, Illinois

Research Focus: Perinatal PCB exposure and acute alcohol challenge in adolescence; immediate early gene (FOS) mapping.

Techniques: cryostat brain tissue, sort and mount tissue, immunohistochemistry (IHC), confocal microscopy and imaging, cell quantification using Image J, and statistical analysis (GraphPad Prism).

Undergraduate Integrative Research Assistant

06.2022–08.2024

PI: Dr. Matt Von Konrat; Head of Botanical Collections; McCarter Collections Manager, Bryophytes and Pteridophytes

Research Focus: Machine-learning pipelines for bryophyte specimen image classification.

Techniques: Data collection, data processing and splitting, model selection/training/evaluation via model retraining and versioning using Python and Jupyter Notebooks.

MANUSCRIPTS

[2] **Gia M Valdez**, Jennifer Dinh, Simone Rhodes, Margaret R Bell. Effects of acute alcohol on adolescent rat brain responses after gestational Polychlorinated Biphenyls exposure (in preparation, 2026).

[1] Margaret R Bell, Katherine A Walker, **Gia M Valdez**, Carissa E Dressel, Advances and challenges in studying effects of EDCs on tissue-resident macrophages in inflammation, *Journal of the Endocrine Society*, Volume 10, Issue 4, April 2026, bvag044, <https://doi.org/10.1210/jendso/bvag044>

POSTERS & PRESENTATIONS

[7] **Gia M Valdez**, Jemimah Ross, Lidan Zhao, Robert Sargis, Margaret R Bell. Effects of Early-Life Exposure to PFAS on Adolescent Neuroimmune Activity. *The Toxicologist*, a Supplement to *Toxicological Sciences*, Abstract Submitted for **Mar 2026**.

[6] **Gia M Valdez**, Jennifer Dinh, Margaret R Bell. Gestational exposure to polychlorinated biphenyls alters adolescent neuroimmune responses to ethanol challenge in the limbic system. *The Toxicologist*, a Supplement to *Toxicological Sciences*, Abstract #4654. **Mar 2025**.

[5] **Gia M Valdez**, Jemimah N Ross, Carissa E Dressel, Margaret R Bell. Effects of early life environmental contaminant exposure on hypothalamic responses to acute alcohol challenge in adolescence. Pr No. 334. 2024 Neuroscience Meeting Planner. Washington, D.C.: Society for Neuroscience, **Oct 2024**.

[4] **Gia M Valdez**, Jemimah Ross, and Margaret R Bell. Effects of Early Life Exposure to Polychlorinated Biphenyls (PCBs) on Neural Responses to Acute Alcohol Intake in Adolescent Rat Prefrontal Cortex. *The Toxicologist*, a Supplement to *Toxicological Sciences*, Abstract #3814. **Mar 2024**.

[3] **Gia M Valdez** and Desiree Valdez. Understanding the basics of Machine Learning Workshop. The Field Museum. Summer Seminar Series. **Aug 2023**.

[2] Dressel CE, **Valdez GM**, Bell MR. SAT427 Effects Of Polychlorinated Biphenyls (PCBs) On Transcription Factor NFkB Activation In Neonatal Rat Microglia Primary Culture. *J Endocr Soc*. 2023;7(Suppl 1):bvad114.1058. Published **Oct 2023**. doi:10.1210/jendso/bvad114.1058

[1] **Gia M Valdez**, Carissa E Dressel, Margaret R Bell. Effects of Early Life Exposure to Polychlorinated Biphenyls (PCBs) on Immediate Early Gene Expression Responses to Acute Alcohol Intake in Adolescent Rats. *The Toxicologist*, a Supplement to *Toxicological Sciences*, Abstract #3944. **Mar 2023**.

HONORS AND AWARDS:

Neuroscience Scholars Program (NSP) Associate Society for Neuroscience	2024-present
Graduate Research Fund recipient College of Science and Health, DePaul University	2024

Title: "Effects of Early Life Exposure to Polychlorinated Biphenyls (PCBs) on Neural Responses to Acute Alcohol Intake in Adolescent Rat Hypothalamus."
 Master's-Undergraduate Scholarly Engagement (MUSE) Award 2024
 College of Science and Health, DePaul University
 Title: "Effects of early-life PCB exposure on Neural and Stress Responses after an Inflammatory Challenge."
 Role: Master Student Mentor
 2024 Perry J. Gehring Diversity Student Travel Award 2024
 Society of Toxicology
 2023 Undergraduate Diversity Travel Award 2023
 Society of Toxicology
 OSSD Undergraduate Attendee Award 2023
 Organization For the Study of Sex Differences
 Undergraduate Research Assistant Program recipient 2022
 College of Science and Health, DePaul University
 Title: "PCB effects on neural and microglial responses to ethanol in adolescent rats"
 Undergraduate Student Research Project recipient 2022
 College of Science and Health, DePaul University
 Title: "Effects of early-life exposure to an environmental contaminant (PCBs) effect on gene expression in response to ethanol within the amygdala"
 Illinois Louis Stokes Alliance for Minority Participation Fellow 2022-23
 College of Science and Health, DePaul University, and The Field Museum
 Title: "From Morphology to Machine Learning: Accelerating Image Classification of Microscopic Organisms"
 Conforti Scholarship recipient 2023
 College of Science and Health, DePaul University
 Given to a promising undergraduate pursuing a degree in STEM.
 DOORS Scholarship recipient 2022
 Promega and Biopharmaceutical Technology Center (BTCI Institute)
 Given to 10 promising undergraduate scientists from underrepresented backgrounds.
 Winners receive USD 5000 for school-related expenses, a Promega mentorship, and a trip to Madison, WI to meet industry leaders, scientists, and mentoring sessions.

TEACHING

Graduate Assistant Mentor DePaul University, Chicago, IL Summer 2025

Summer Research Program: AI for Healthcare

Role: Guided interpretation of Python/R model outputs, facilitated hands-on exercises, and advised student presentations.

Teaching Assistant DePaul University, Chicago, IL

BIO 260: Genetics Spring 2025

Role: Guided students through fly characteristics and linkage mapping; taught Hardy-Weinberg labs; taught chi-square and basic biostatistics.

BIO 250: Cell Biology Fall 2024, Winter 2025

Role: Guide experiments in protein/enzymatic assays (SDS-PAGE); prepared and graded quizzes and exams; held twice-weekly tutoring sessions on class/lab material.

BIO 202: Physiology Winter 2023-2024
Role: Led ECG, blood pressure, and spirometry labs; created targeted review sessions linking lab exercises with class content; graded lab and exam materials.

BIO 201: Anatomy Fall 2023, Spring 2023
Role: Directed labs using human models; integrated histology to connect gross and tissue structure; managed lab setups and created study guides.

PROFESSIONAL AFFILIATIONS

Society for Neuroscience	2024
Organization for the Study of Sex Differences	2023
Society of Toxicology	2023
Japanese Language Honors Society	2022
Nu Rho Psi (NPΨ) Honors Society	2022
Psi Chi Honors Society	2021

SERVICE

Registration Volunteer	2023- 25
Japanese American Society (JASC), Chicago, IL	
Poster Reviewer	2023
Undergraduate STEM Research Showcase, DePaul University, Chicago IL	
Speaker for “Strategies for Success in CSH”	2023
College of Science and Health, DePaul University, Chicago IL	
Conference Volunteer	2023
International Society of Exposure Science (ISES), Chicago IL	
Treasurer and Community Activities Coordinator	2022-23
Neuroscience Club, DePaul University, Chicago IL	

SKILLS

Languages: Conversational level in Japanese, French, and Spanish
Programming: Proficient in R, Python, Jupyter Notebooks, and SQL
Statistical Programs: Proficient in SPSS, GraphPad Prism, and Jamovi